Prodigy Anesthesia Quick Review

Diabetes Mellitus

What is the difference between type I and type II diabetes?	Type I diabetes is caused by an immune-related destruction of beta cells in the pancreas. Type II diabetes is not immune-related and results from a deficiency of insulin and/or a defect in insulin receptors. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 376-377.*
What percent of patients with diabetes mellitus are type I?	5-10% *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 376.*
What are the acute complications that occur due to diabetes mellitus?	Hypoglycemia, diabetic ketoacidosis, and hyperglycemic nonketotic coma. *Fleisher LA. Anesthesia and Uncommon Diseases. 6th ed. Philadelphia, PA: Elsevier Saunders; 2012: 428-429.*
What immune cell is implicated in the autoimmune destruction of beta cells in type I diabetes?	The destruction of beta cells is mediated by T cells although the exact mechanism that triggers the destruction is unknown. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 376-377.*
What lab values are diagnostic of type I diabetes?	A random blood glucose greater than 200 mg/dL and a hemoglobin A1C greater than 7.0% *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 377.*
How long after the onset of type I diabetes does it take before the onset of symptoms?	Clinical symptoms of diabetes may not present for as long as 10 years. About 80-90% of beta cells must be destroyed before hyperglycemia will appear. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 377.*
What is the function of insulin?	Insulin facilitates the transport of glucose and potassium into the cell, and is important for the cellular uptake of glucose with the exception of the brain and liver where it does not affect glucose transport. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1343.*
How is insulin metabolized?	Insulin is metabolized by both the liver and kidneys *Barash PG, Cullen BF, Stoelting RK. Clinical Anesthesia. 5th ed. Philadelphia, PA: Lippincott Williams and Wilkins: 2006: 1145.*

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What is the normal insulin prodution in a day?	Normal insulin production in the adult is about 40 to 50 units per day *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1343.*
What forms of insulin are considered short-acting?	Human regular, Lispro (Humalog), and Aspart (Novolog). Human regular insulin lasts about 5-8 hours and Humalog, and Novalog both last about 3-6 hours. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 380.*
What forms of insulin are considered intermediate-acting?	Human NPH and Lente which both last about 10-20 hours *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 380.*
What forms of insulin are considered long-acting?	Ultralente which lasts 24-48 hours and glargine (Lantus) which lasts about 24 hours. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 380.*
How do alpha-glucosidase inhibitors help control blood glucose levels?	Alpha-glucosidase inhibitors decrease postprandial glucose absorption. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1344.*
How do sulfonylureas help control glucose?	Sulfonylureas increase insulin secretion by beta cells. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 857.*
How do thiazolinediones and metformin help control blood glucose levels?	Thiazolinediones and metformin (a biguanide) enhance tissue sensitivity to insulin *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1344.*
What are the symptoms of hypoglycemia?	Hypoglycemia is associated with a sympathetic response and symptoms such as hypertension, tachycardia, diaphoresis, and lacrimation. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 860.*
How do elevated glucose levels cause end-organ damage?	Vasodilation caused by hyperglycemia prevents the organs from protecting themselves from hypertension by increasing systemic vascular resistance in organ vasculature. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 856.*

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What percent of type I and type II diabetics develop end-stage renal disease?	40% of type I diabetics and about 5% of type II diabetics develop end-stage renal disease. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 383.*
What is the progression of symptoms as renal disease associated with diabetes develops?	The symptoms begin with hypertension, albuminuria, and peripheral edema, and progress to a decrease in glomerular filtration rate due to glomerulosclerosis and tubulointerstitial disease. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 383.*
What is the chief indicator that diabetes associated renal impairment has progressed to advanced glomerulonephritis?	Patients can be asymptomatic for as long as 15 years, but the appearance of proteinuria is the hallmark sign the glomerulosclerosis is severe. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 383.*
What percentage of diabetics will develop peripheral neuropathy?	About 50% of patients with diabetes over 25 years will develop peripheral neuropathy. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 383.*
How does diabetic retinopathy occur and how can it be prevented?	Diabetic retinopathy occurs as a result of occlusion and dilation of retinal arterioles, microvascular aneurysms, hemorrhages, and fibrotic changes. Strict maintainance of glucose within normal ranges helps prevent these alterations. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 383.*
How does hyperglycemia affect connective tissue?	Elevated glucose levels promote glycosylation reactions that decrease elasticity in connective tissue resulting in stiff-joint syndrome and atlanto-axial stiffness. This can also lead to impaired wound healing. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 856.*
How does blood glucose affect global ischemia?	Studies have correlated elevated glucose levels with poor short-term and long-term outcomes in patients with brain damage from global ischemia. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 710.*
What are the symptoms of autonomic neuropathy due to diabetes and why is this important?	Signs of autonomic neuropathy include a lack of orthostatic changes in the heart rate, early satiety, lack of sweating, and impotence. Patients with autonomic neuropathy are at increased risk for gastroparesis and silent myocardial infarction. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 856.*

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How can serum glucose levels affect serum sodium concentration?	For each 100 mg/dL increase in the serum glucose concentration, you would expect to see a corresponding 1.6 mEq/L decrease in the serum sodium concentration. As the increased glucose level increases the plasma osmolality, water is drawn into the vascular space, which dilutes the serum sodium concentration. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 1114.*
What are the symptoms of hyperglycemic hyperosmolar syndrome?	Polyuria, polydipsia, hypotension, tachycardia, hyperosmolarity (>340 mOsm/L) and hypoperfusion of major organs. Intravascular coagulation and mesenteric thrombosis are a significant risk in hyperglycemic hyperosmolar syndrome. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 382.*
What are the characteristics of hyperglycemic hyperosmolar syndrome?	Patients with HHS exhibit exaggerated hyperglycemia, hyperosmolarity and dehydration. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 382.*
What are the diagnostic features of hyperglycemic hyperosmolar syndrome?	Glucose levels > 600 mg/dL, a pH > 7.3, a serum bicarbonate > 15, and a serum osmolarity > 350. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 382.*
Is hyperglycemic hyperosmolar syndrome more common in type I or type II diabetics?	In contrast to diabetic ketoacidosis which occurs more commonly in type I diabetics, hyperglycemic hyperosmolar syndrome is more common in type II diabetics. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 383.*
What are the diagnostic features of diabetic ketoacidosis?	Patients in diabetic ketoacidosis exhibit a serum glucose > 300 mg/dL, hypovolemia, hyponatremia, a pH < 7.3, a serum bicarbonate level < 18, an osmolarity < 320, and a high level of ketones in the blood. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 382.*
Over what period of time does hyperglycemic hyperosmolar syndrome occur?	It typically occurs over days to weeks. It usually develops in elderly patients who are isolated and experience a stressful medical event and are unable to recognize symptoms or have them recognized by another person. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 382.*
What electrolye abnormalities usually occur with diabetic ketoacidosis?	Hyperglycemia results in hyponatremia, hypophosphatemia, and hypokalemia, and hypomagnesemia. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 382.*

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What is the treatment for diabetic ketoacidosis?	The treatment of DKA is comprised of hydration with large quantities of normal saline, administering insulin, and replacing lost electrolytes. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 382.*
What factors may trigger diabetic ketoacidosis in type I diabetics?	Stressors such as infection, myocardial infarction, acute pancreatitis, and stroke. Omission of regularly scheduled doses of insulin are responsible for 15-20% of cases of DKA. DKA can also occur as the presenting symptom of type I diabetes mellitus. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 382.*
Does diabetic ketoacidosis occur more commonly in patients with type I or type II diabetes mellitus?	Diabetic ketoacidosis occurs more commonly in type I diabetes mellitus. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 382.*

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